③ penicilline V is an antibiotic with config 2S, 5R, 6R; but its enantiomer which does not occur naturally has no antibiotic activity.

17.06.25

CHAPTER 02

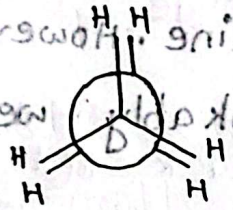
Configuration & Conformation of cyclo compounds

6-membered cyclohexane is most stable.
6 এর বেশি হলে strain এর জন্য unstable হয়।

△ □ ⬠ ⬡ → sp^3 hybridized

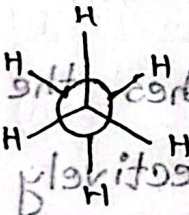
Conformation of cyclopropane :

Cyclopropane suffers from both angle & torsional strain. Each -C-C- bond is eclipsed due to rigidity of the molecule. Three bond angle (60°) are sp^3 hybridized.



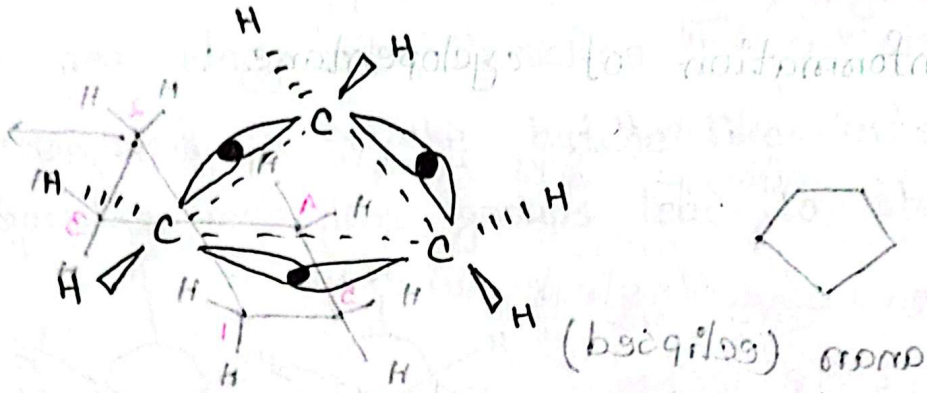
এরকম eclipsed form এ থাকায় H–H এর repulsion এর কারণে তারা bond rotation করে staggered form এ আসতে চায়। এটা torsional strain





আবার sp^3 এর bond angle 109.8° না হয়ে 60° হয়, তাই এতে bond angle strain-ও highest থাকে।

The highest electron density of the C-C bonds does not lie along lines connecting the C-atoms. Bonding electrons lie principally outside of the triangular internuclear lines resulting in bend bonds.



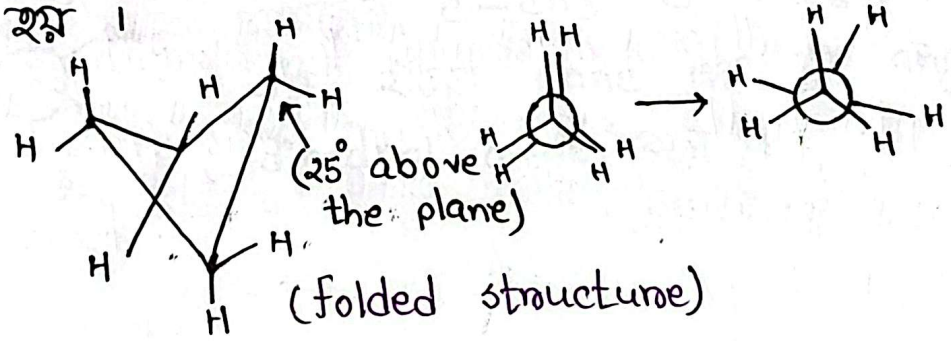
Conformation of cyclobutane : 90° ; square planar

Angle strain : bond angle এখানে 90° হওয়ায় angular strain cyclopropane থেকে অনেক কম।

$(109 - 90)^\circ$

Torsional strain : 8 টা H এর জন্য এর torsional strain অনেক বেশি।

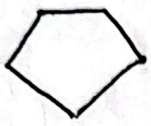
Angle strain এর জন্য square planar structure হতে পারলেও T.S এর জন্য এই structure হতে পারে না, 2টাই balance করার জন্য ১টা নতুন folded structure তৈরি হয়।



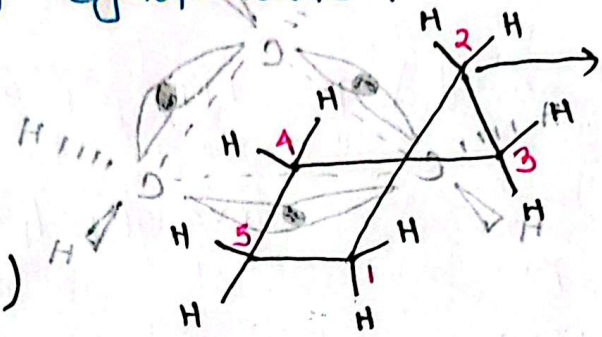
(folded structure)

এইটায় angle strain হালকা বেড়ে গেলেও torsional টা অনেক কমে যায় ও ২টাই মোটামোটি stable হয়, তাই for 8টা H eclipsed থেকে একটু দূরে থাকে তাই T.S. কমে। তাই folded বেশি favourable.

Conformation of cyclopentane :



planar (eclipsed)



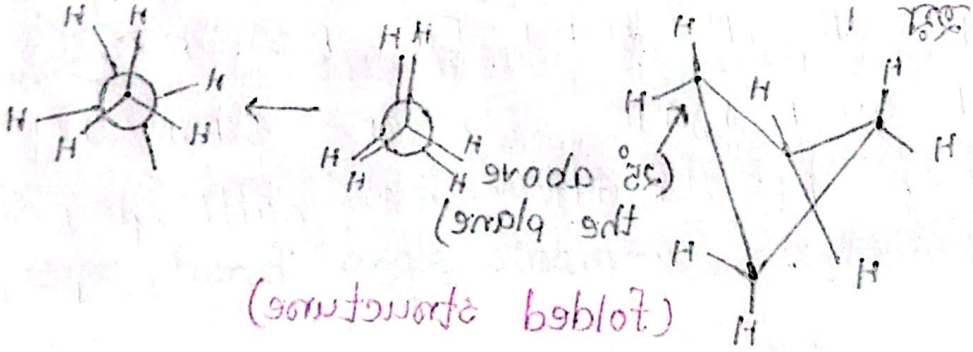
(out of plane, methylene group)

envelop / Bent shape

Angular strain : 108° ; তাই angle strain is almost zero

Torsional strain : 10টা H এর জন্য T.S অনেক বেশি হয়।

Bent shape - এ H গুলা একটু দূরে থাকায় T.S. অনেক কমে যায়। তাই এই shape more favourable



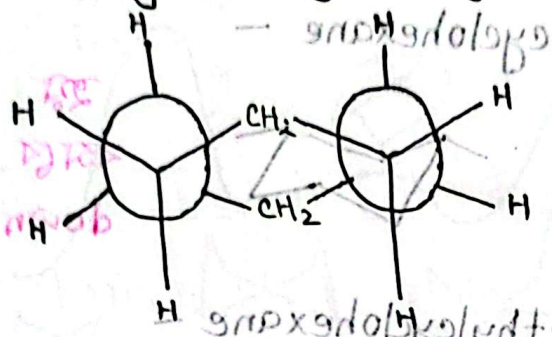
18.06.25

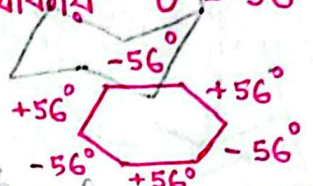
Conformation of cyclohexane :

mainly chain conformation.

① Cyclohexane molecule exists predominantly in non-planar, puckered conformation ; the chain conformation.

② All bonds are staggered, therefore pitzer strain is minimized. However, a gouche butane like interaction betⁿ neighbouring methylene groups lead to steric strain.

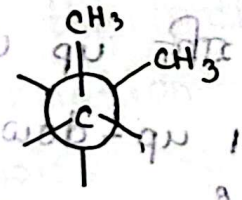




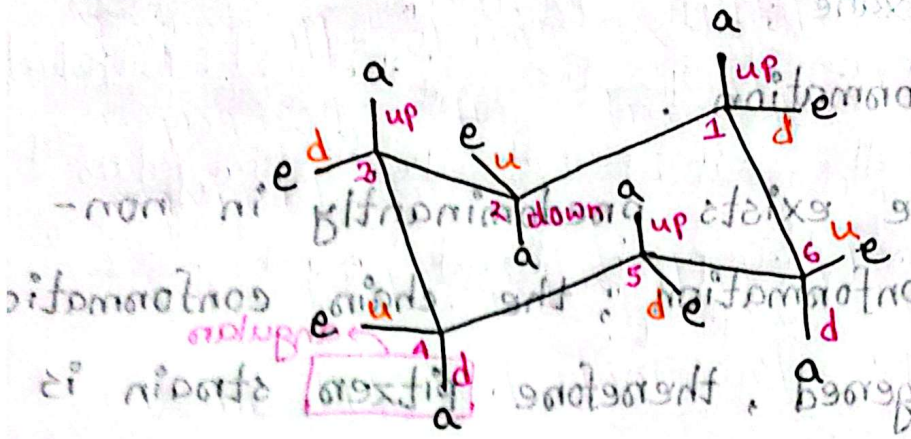
③ The dihedral angles θ are not exactly 60°, but these are around 56°.

④ Each ring, C–C bond is labelled with a number corresponding to dihedral angle and the sign corresponding to clockwise (+) or anticlockwise (−) for the angle.

⑤ The bond angles are not exactly 109.3° but 111°.



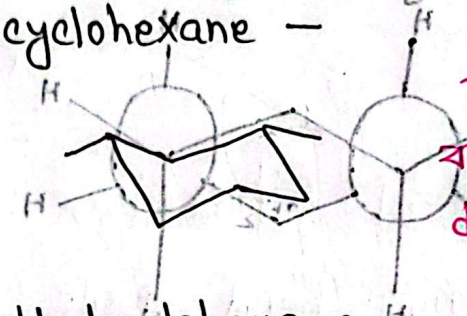
সামনের $C-CH_3$ ও পিছনের $C-CH_3$ — এই দুটার plane এর যে angle, তাকে dihedral angle বলে।



a = axial
e = equatorial

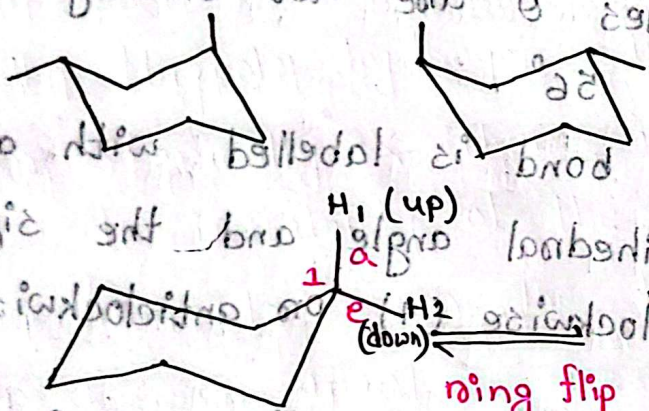
a/e always, এটা পরপর up / down হয়।

cis-1,3-dimethylcyclohexane —



হয় দুটাই up-এ বসবে, নাহয় দুটাই down

trans-1,3-dimethylcyclohexane —



ring flip হলে axial → equatorial
& e → a হয়ে যায়। কিন্তু H, যদি up এ থাকে, flip এর পরেও up-এই থাকবে। up-down এ H এর position একই থাকবে, শুধু a ↔ e হয়ে যাবে।

Temp যখন খুব কম (-80°/-120°) হয়, তখন ring flip খুব slow হয়। তখন NMR এর সাহায্যে সহজেই axial/equatorial hydrogen differentiate করা যায়। (Room temp -এ flip easily হয়।

* Conformational analysis of cyclohexane :

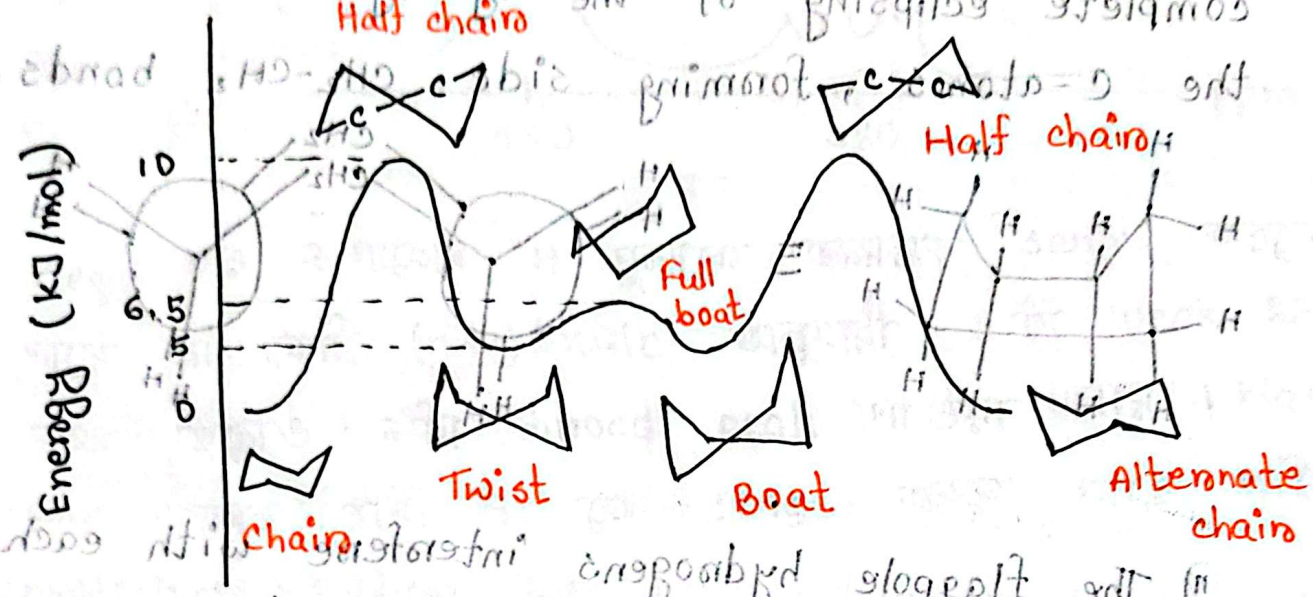


Fig: The activation energy for cyclohexane in ring inversion

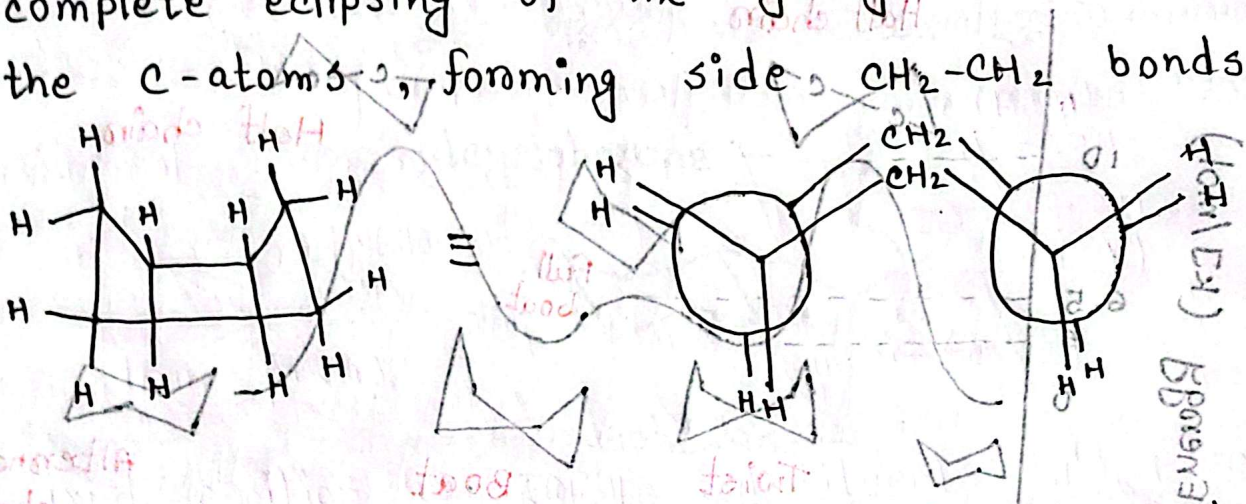
One chair conformation of cyclohexane can be easily converted into an alternate chair conformation state along the path of one chair to the alternate chair conformation provides another example of conformational analysis.

The two chairs & the twisted boat conformations are intermediates represents every minima & the

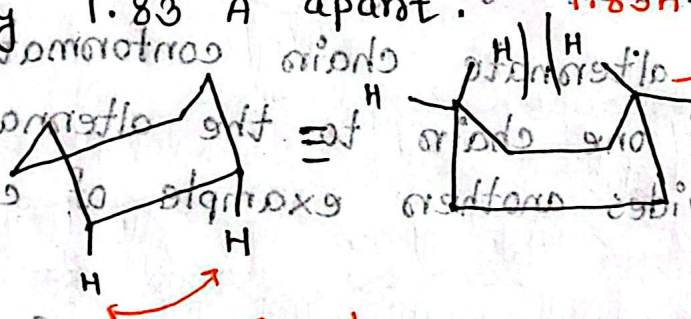
two half chains & full boat conformation represents the transitional state (energy maxima).

The boat conformation of cyclohexane:

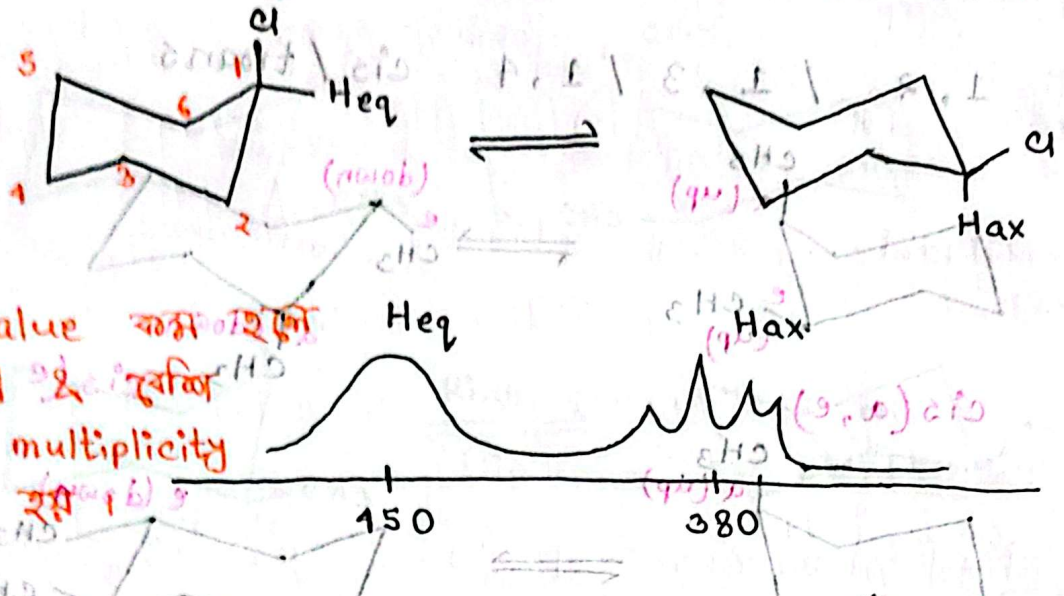
i) In the boat form of cyclohexane, there is complete eclipsing of the hydrogens attached to the C-atoms, forming side CH_2-CH_2 bonds.



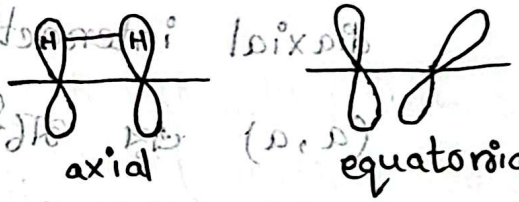
ii) The flagpole hydrogens interfere with each other sterically & there is a 2nd factor which leads to instability of the boat form in comparison to chair. The flagpole hydrogens are only 1.83 Å apart.



* Conformation of chlorocyclohexane at -150°C:



Heq তার ২ পাশের H গুলোকে ভালমত sense করতে পারে না তাই (n+1) rule অনুযায়ী 5 টা peak থাকার কথা হলেও ১টা broad peak পাওয়া যায়। Hax তার পাশের সব H কে sense করতে পারে তাই multiplicity show করে।



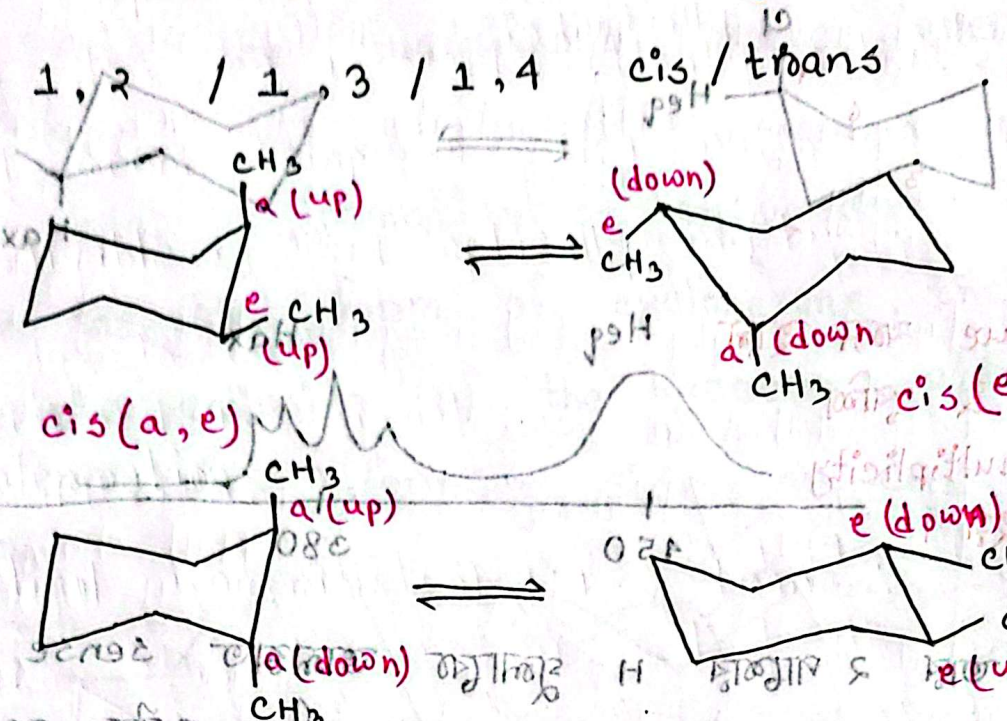
cyclohexane এ covalent bond এর e⁻ এর জন্য electric field তৈরি হয়।

The equatorial proton is coupled with neighbouring protons which are also gauche with lower coupling constant value. The axial proton is coupled with both equatorial / axial hydrogens with high coupling constant & gives a well resolved multiplicity.

01.07.2025

* Conformation of disubstituted cyclohexane:

1,2 / 1,3 / 1,4 cis / trans

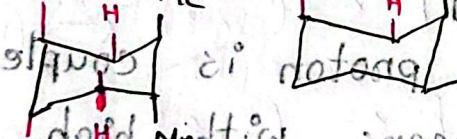


trans (a, a)

trans (e, e)

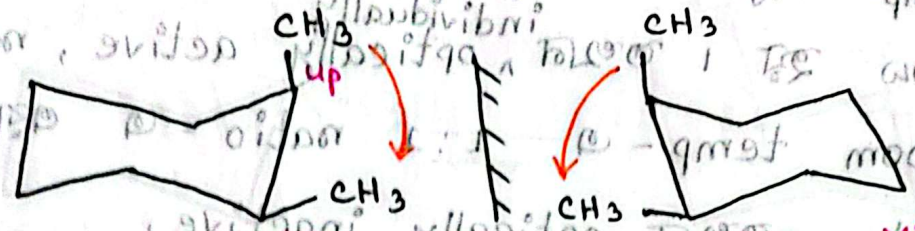
axial position এ কোনো substituent আসলে তা diaxial interaction এর জন্য unstable

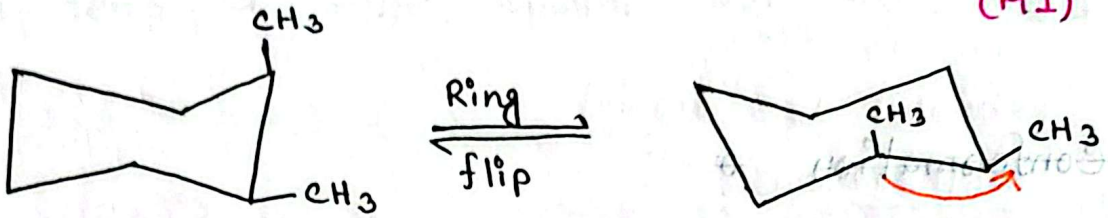
(a, a) এর মধ্যে total two axial interaction থাকে। (a, e) & (e, a) তে ১টা করে। (e, e) এ এমন কোনো interaction থাকে না তাই এটা most stable.



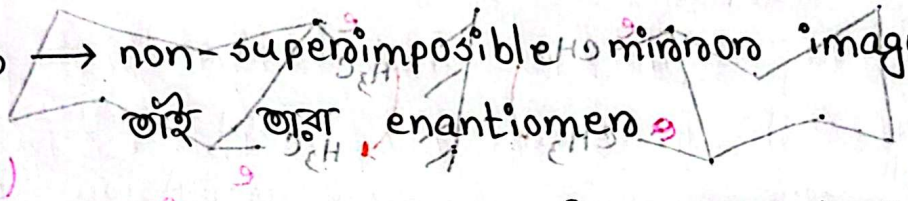
Stability order: (e, e) > (a, e) ≈ (e, a) > (a, a)

Optical isomerism of 1,2-cis dimethyl cyclohexane :





A & B → non-superimposible mirror image
তাই এরা enantiomer



B & C same বা ⁴ হলে configurational isomer & same হলে conformational isomer.

A → a → e — up-up clockwise

B → a → e — up-up anticlockwise

C → a → e — up-up anticlockwise

A যদি a → e notation ধরি, বাকি দুলোও a → e ধরতে হবে

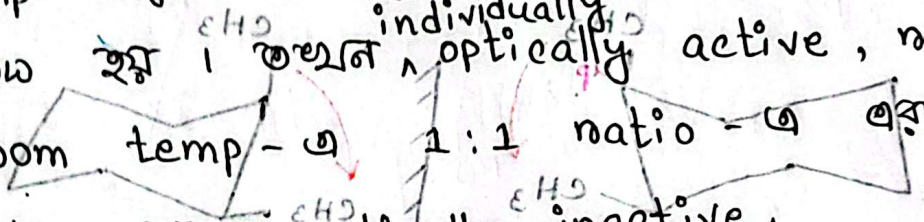
Hence, B & C are same.

Conformational enantiomers; individually → optically active.

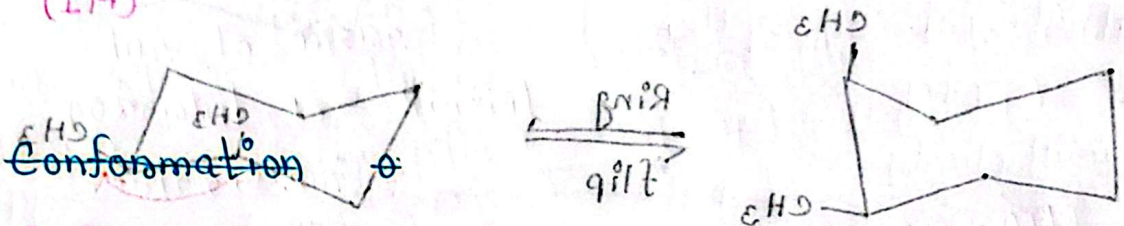
→ non-resolvable

1 : 1 racemic mixture → optically inactive

যদি temp অনেক কমিয়ে দেওয়া হয়, তখন ring flip slow হয়। তখন individually optically active, resolvable. কিন্তু room temp-এ 1:1 ratio-এ থাকা non-resolvable. তখন optically inactive.

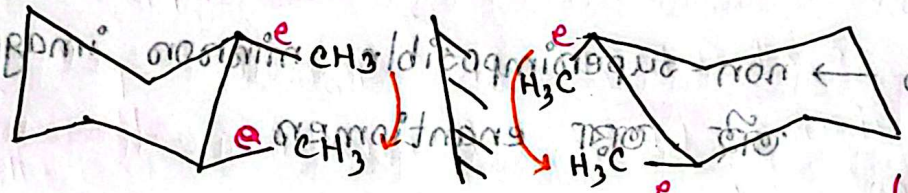


(IH)



Optical isomerism of 1,2 trans :

(IA)

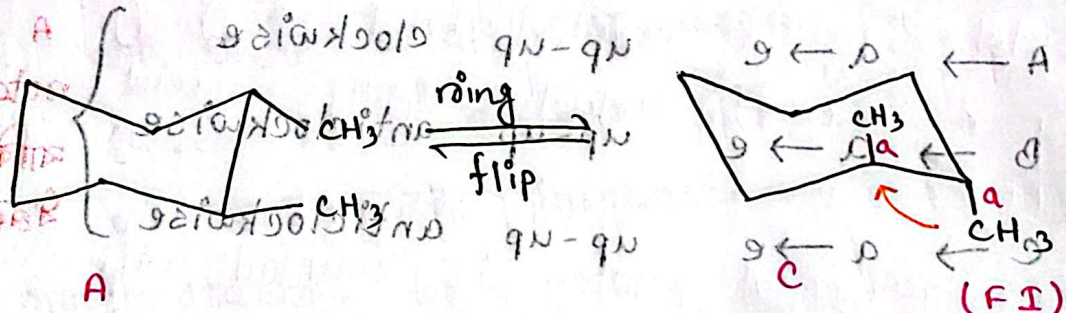


(MI)

A

B

Enantiomers



(FI)

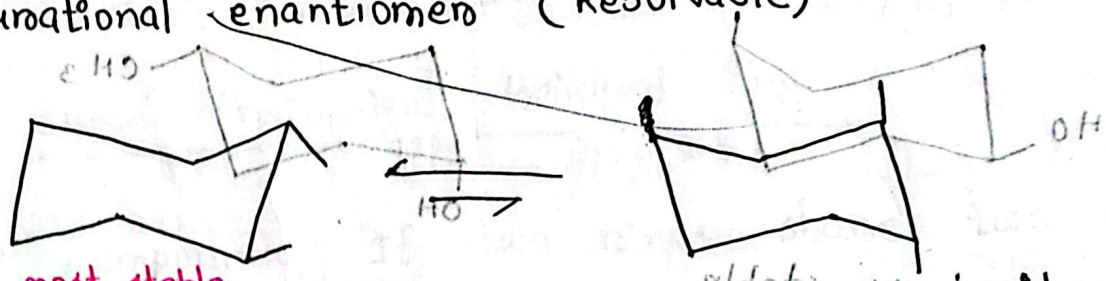
A → (e,e) → d, u → clockwise

B → (e,e) → d, u → anti

(a,a) → d, u → clockwise

Hence, B & C are not same. তাই তারা

Configurational enantiomer (Resolvable)



most stable

Room temp এ এদের আলাদা করা যাবে easily.

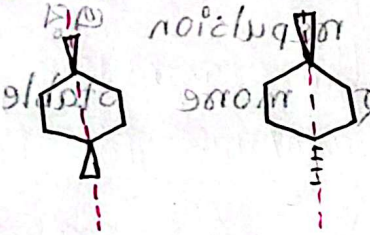
* cis-1,3-dimethyl cyclohexane :



i) Have plane of symmetry [C-C cut করা হয়]

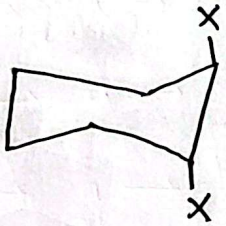
তাই optically inactive

cis-1,4-dimethyl cyclohexane :



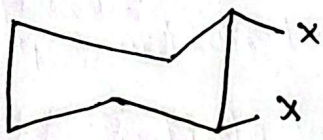
plane of symmetry আছে।

*



(a,a)

More stable

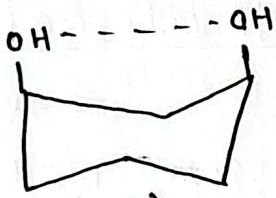


(e,e)

less stable due to dipole dipole repulsion

X = Halogen

যৌগটা (a,a)-এ 1,3 diaxial interaction এবং dipole repulsion থাকায় dipole repulsion prominant



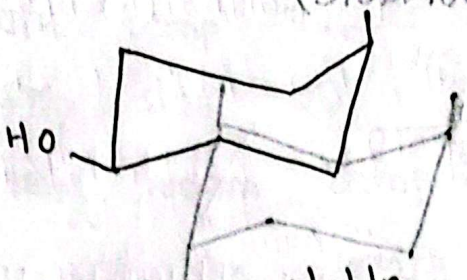
(a,a)

More stable due to intramolecular H-bonding

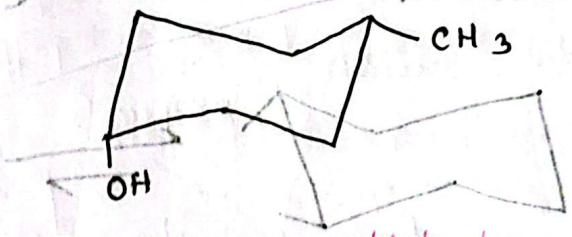


(e,e)

less stable



Less stable due to more powerful 1,3 diaxial interactions

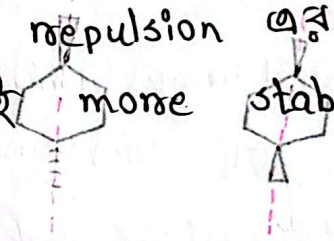


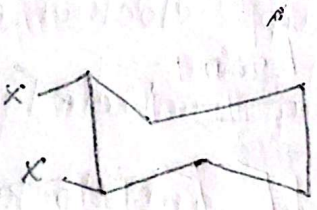
More stable

- OH more electroneg. তাই যখন axial-এ থাকে তখন C – OH bond length বড় হয়।

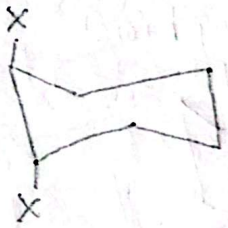


ফলে তার পাশের 1,3 - diaxial interaction তেমন prominent হয় না, repulsion এর জন্য reach করতে পারে না। তাই more stable.

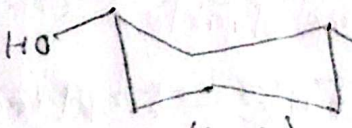




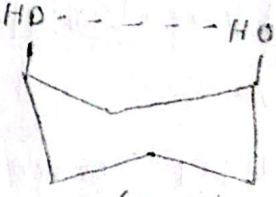
(e, e)



(a, a)



(e, e)

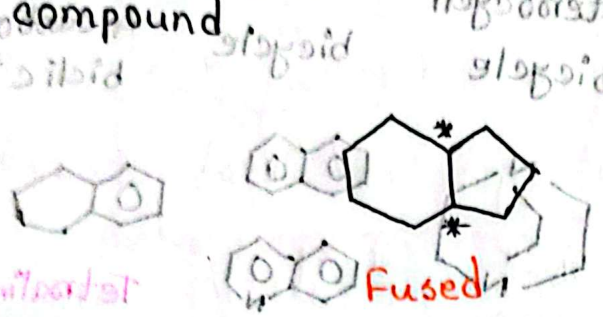


(a, a)

CHAPTER -6

Fused rings and Bridged Rings

Bicyclic compounds : If two rings share two or more common atoms, the compound is called a bicyclic compound.

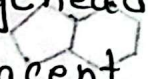




* bridgehead carbon

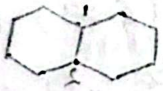


i) Fused bycyclic compound : When the bridgehead carbons of a bycyclic compound are adjacent to each other than the compound is called fused bycyclic compound.



ii) Bridged bycyclic compound : When the bridgehead atoms are not adjacent, then it is called bridge bycyclic compound.



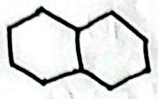




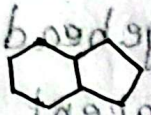
* Classification of Bicyclic compounds :

Bicyclic Compounds

- Carbocyclic bicyclic
- Aliphatic bicyclic
- Heterocyclic bicyle
- Aromatic bicycle
- Aromatic & carbo-bicic



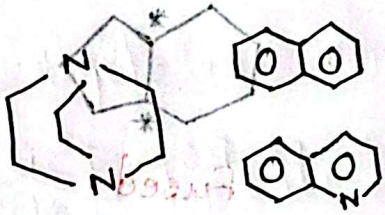
Decaline



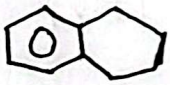
Hydrindane



Bornane



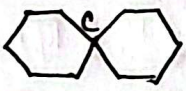
DABCO

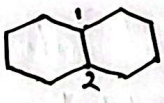


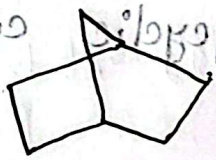
Tetraline

Modes of Joining

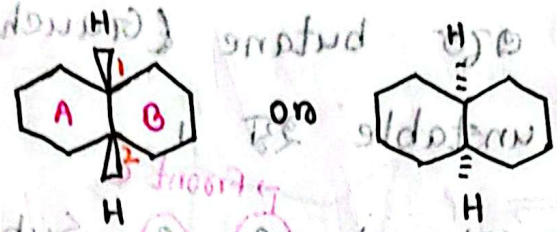
- Spiro
- Fused
- Bridged



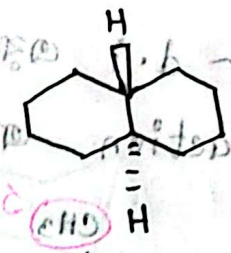


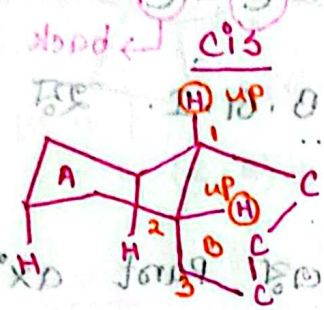


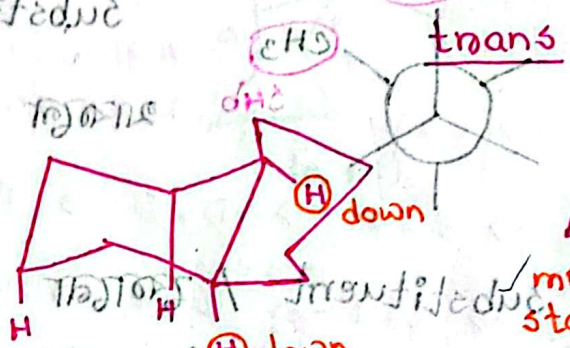
* Decaline :

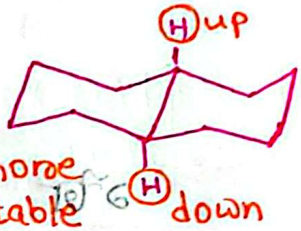


or

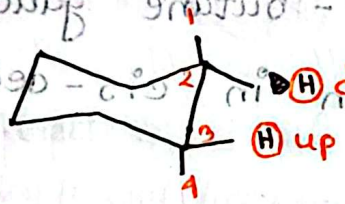




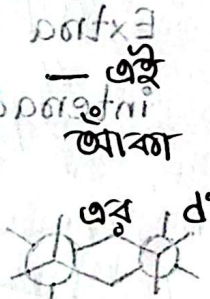




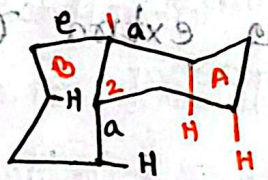
Trans - compound টা বেশি stable কারণ এতে 1,3-diaxial interaction নেই।



— এই position-এ 2nd cyclohexane আসা possible না কারণ 1 ও 1 এর distance অনেক বেশি।



Stability Order :



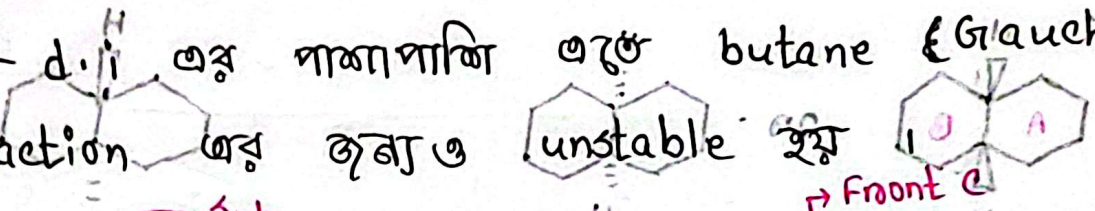
→ four (1,3) diaxial interaction

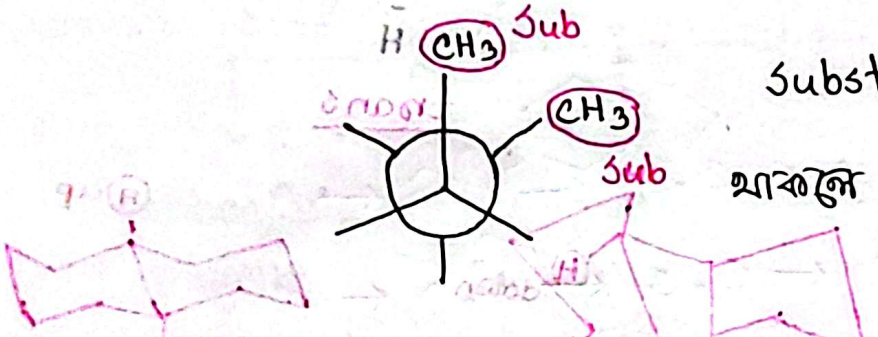
এটা axial-এ ~~এটা~~ substituent আসলে তার জন্য এটা (1,3)-diaxial interact হয়।

A এর 2 এ axial-এ cycle এর CH_2 আছে, সেটা এটা (1,3)-diaxial interaction হবে। আবার B-এর 1-এ axial এ B A-cycle এর CH_2 আছে। তখন সেটা B-cycle এর এটা H এর সাথে (1,3)-d.i face করবে।

4টা (1,3) - d.i থাকায় ⊕ cis - decaline unstable.

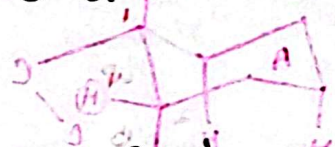
(1,3) - d.i এর পাশাপাশি এতে butane (Gauche interaction এর জন্যও unstable হয়।





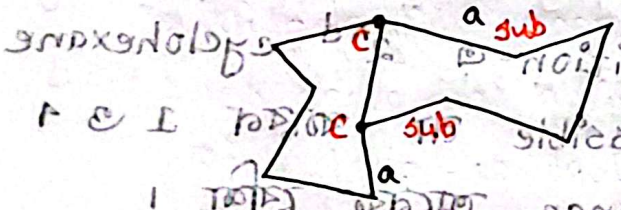
Substituent - C - C - Subs
Front C
back C

থাকলে এই G.G.I. হয়।

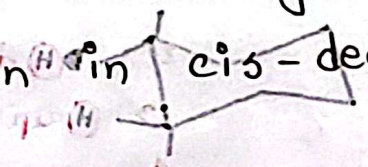


2টা substituent / কোনো cycle এর Pant axial-এ এসে যুক্ত হলে 2টা G.G.I. হয়।

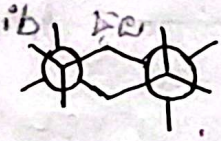
G.G.I থাকলে molecule unstable হয়।



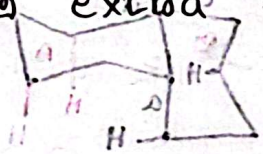
Extra 3 - butane gauche interaction in cis - decaline



প্রত্যেক cis/trans → এ 6+6 = 12টা



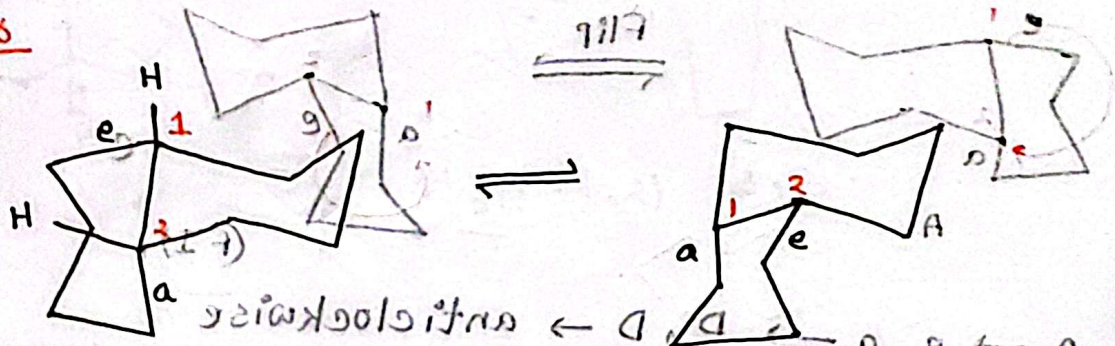
GGI থাকে। cis-decaline এ extra -3টা থাকায় বেশি unstable. trans-decaline এ extra



GGI না থাকায় stable

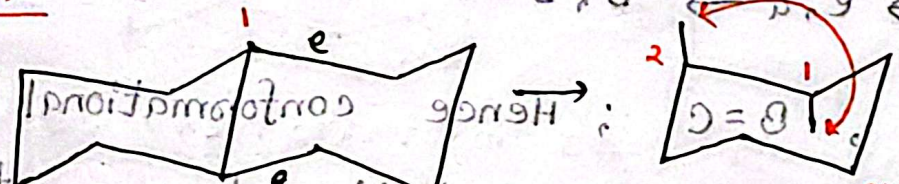
* Flipping of Decaline :

cis



(e, a) ⟶ (a, e)

trans

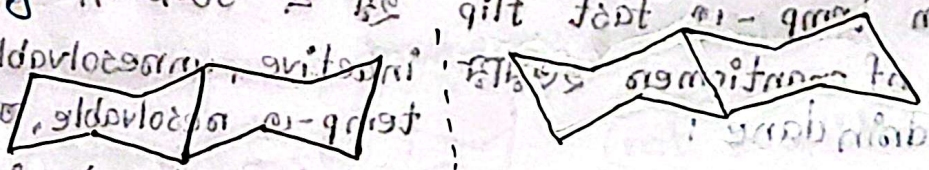


Distance বৃদ্ধি হওয়ায় trans আর flip হয়না

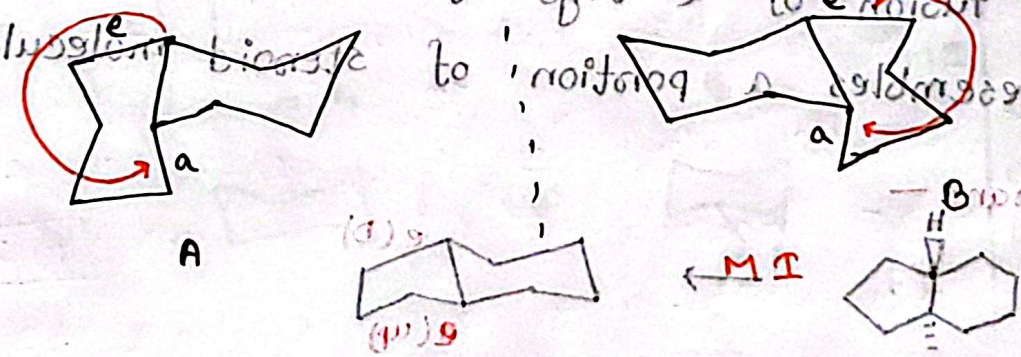
(Not possible)

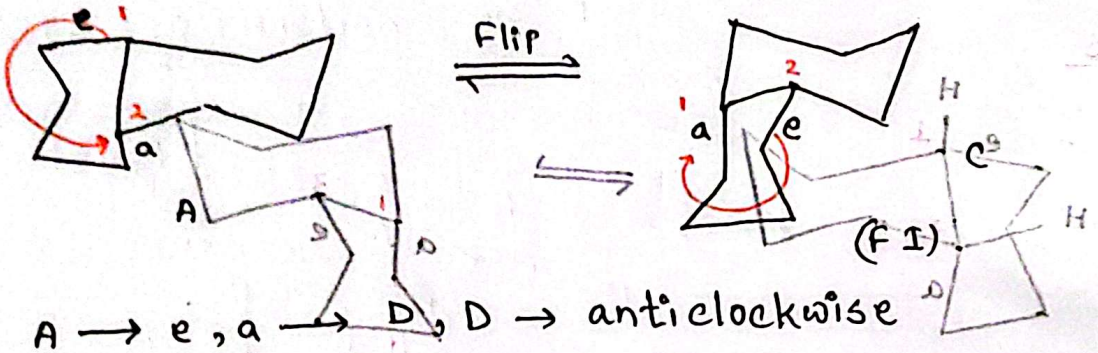
(e, e) (a, a)

* Chirality - in cis / trans Decaline



trans Decaline optically inactive & meso-compound. Plane of symmetry আছে বলে optically inactive





A → e, a → D, D → anticlockwise

B → e, a → D, D → clockwise

C → e, a → D, D → clockwise

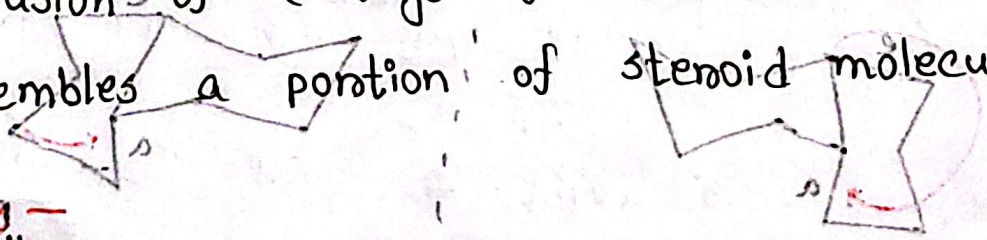
A ≠ B, B = C ; Hence conformational ~~is~~ enantiomer ; Non-resolvable at room temp.

* cis-decaline room temp-এ optically inactive but low temp-এ active কেন ?

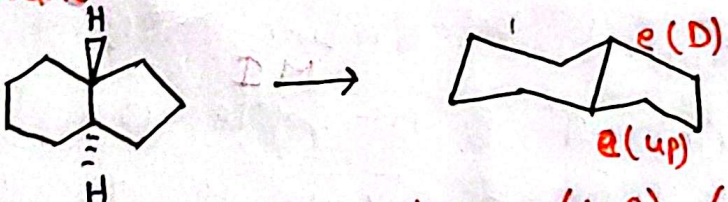
Room temp-এ fast flip হয় & 50% A-C পাওয়া যায়, Mix of enantiomer হওয়ায় inactive, unresolvable. কিন্তু low temp-এ resolvable, তাই active.

* Hydrindane :

Hydrindanes (Octahydro-1H-indane) is an interesting molecule because it results from the fusion of 2 rings of different size & also H resembles a portion of steroid molecule.

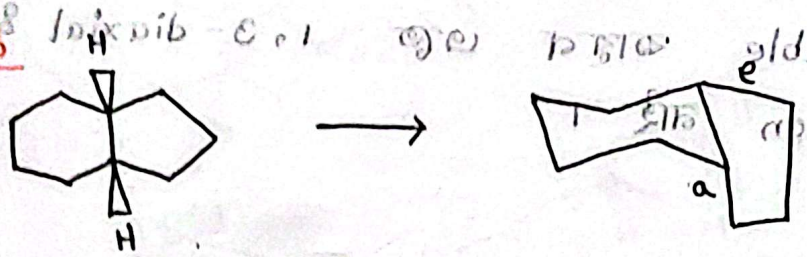


trans —

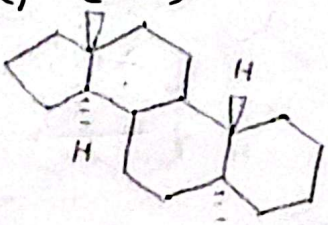


trans (1,2) (e, e)

cis

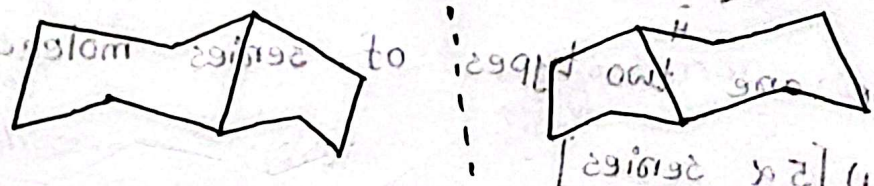


cis (1,2) (e, a)



* Chirality in Hydrindane :

trans :



MI

trans - Hydrindanes are conformationally rigid molecule & occurs as 2 stable enantiomers

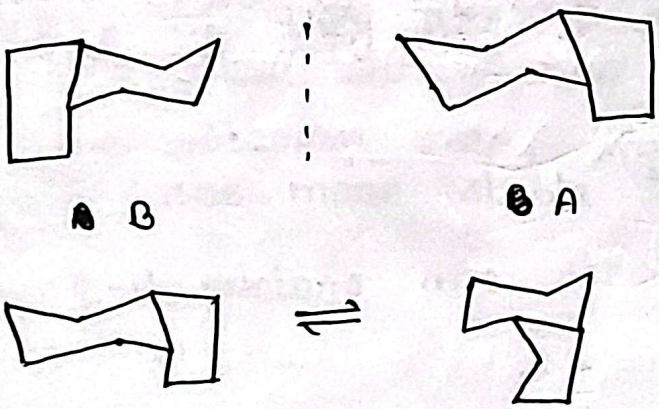
No flipping possible

So configurational isomers ; Resolvable at room temp.



(not possible)

cis :



MI

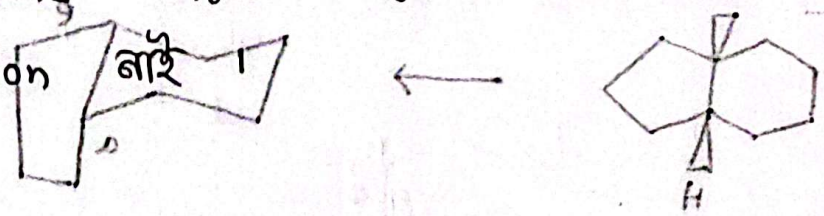
$A \neq B$

$B = C$

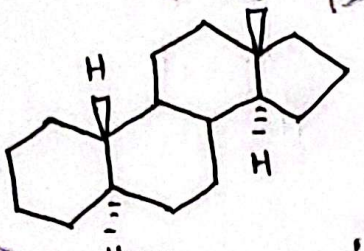
Conformational enantiomers ; Re Non-resolvable at room temp

FI

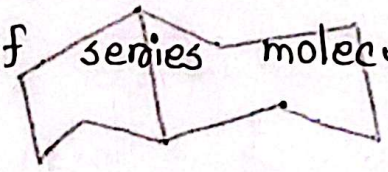
trans - বেশি stable কারণ এতে 1,3- diaxial & gauche interaction নাই।



* Steroid :



There are two types of series molecule :

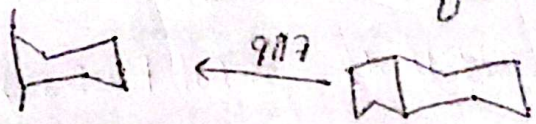


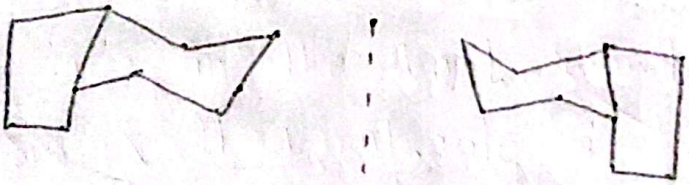
1) [5α series]

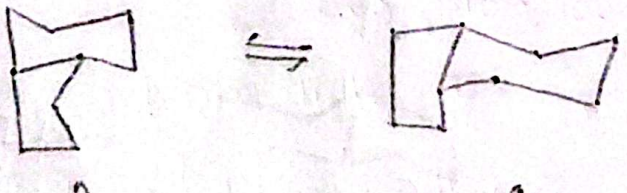
Ring A, B are fused in trans form, B-C trans,

II) [5β series]

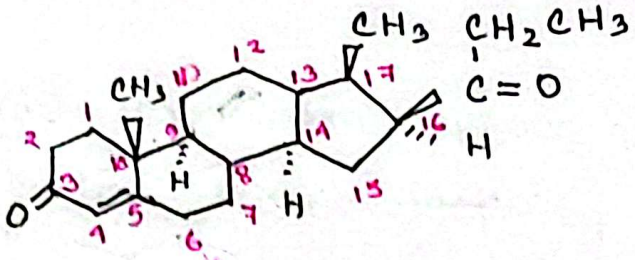
Ring A, B in cis form



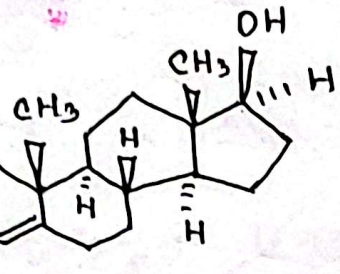




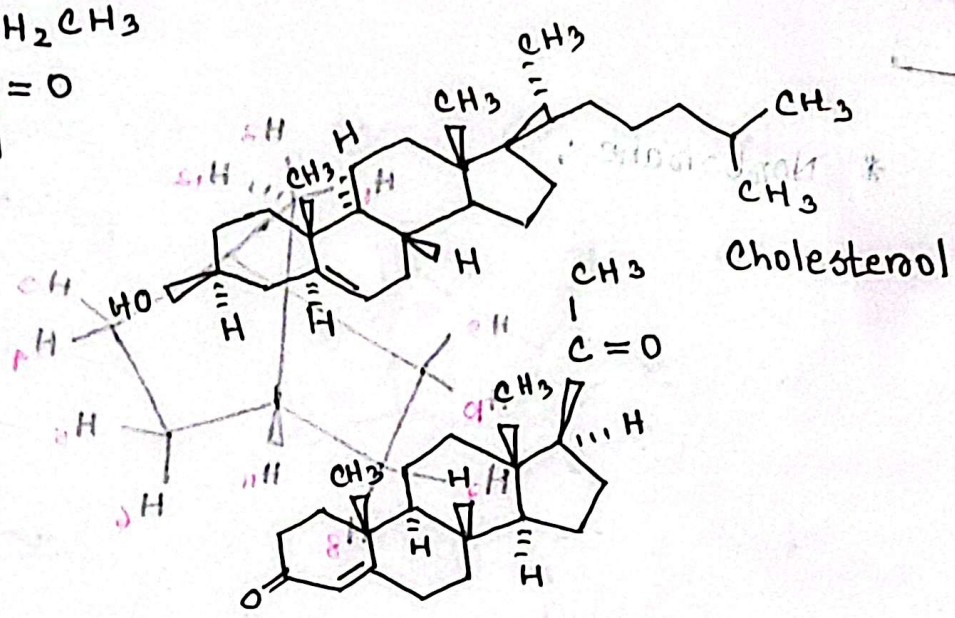
1)



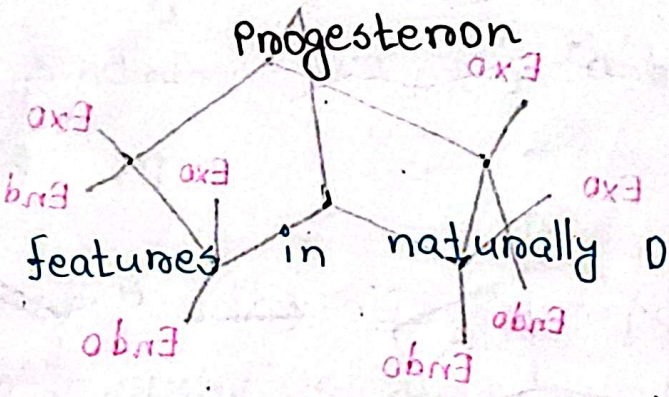
Contison



Testosteron



Cholesterol

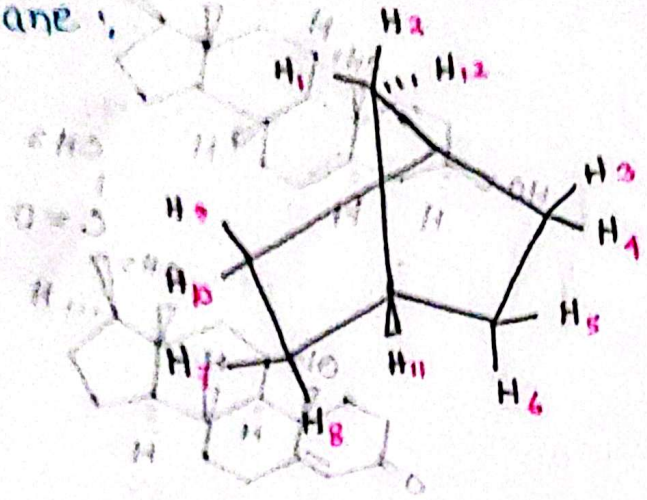


Progesteron

* Common structural features in naturally occuring steroids :

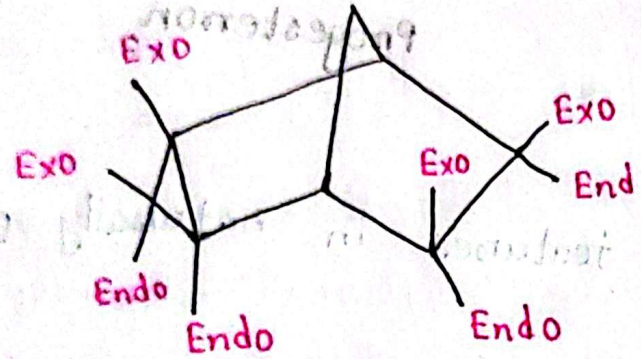
1. All ring fusions are trans because of stability.
2. Many steroids have methyl groups so called angular methyl at 10 & 13 position.
3. Group on atoms which are same as 10-CH_3 & 13-methyl like up they are β & pointing downward (α)
4. 5α series are more stable than 5β.
5. C-1 & C-11 protons are sterically hindered.

* Norbornane :



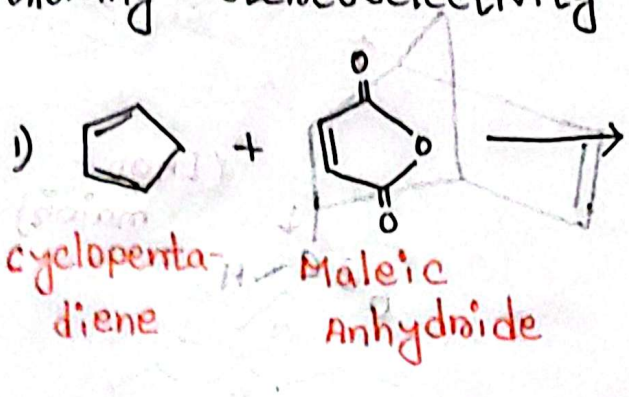
≡



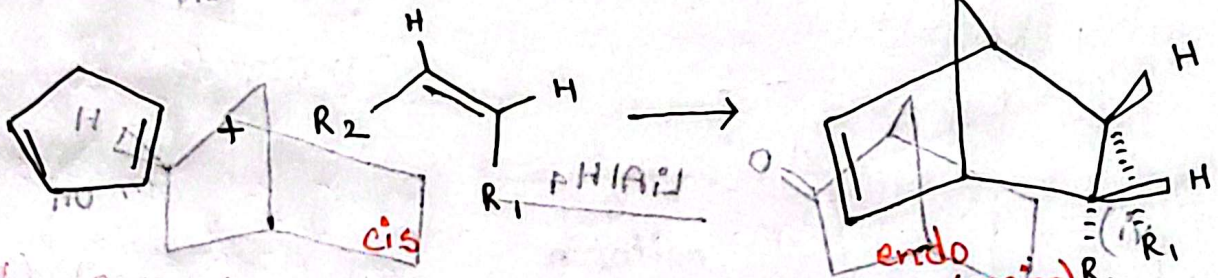


The almost ~~on~~ common bridged bicyclic system is bicyclo [2,2,1] heptane or norbornane. In norbornane C-1 & C-4 are the bridging carbon forcing the molecule to adopt a boat conformation, two butanes are in the boat which are in eclipsed form & the molecule is strained.

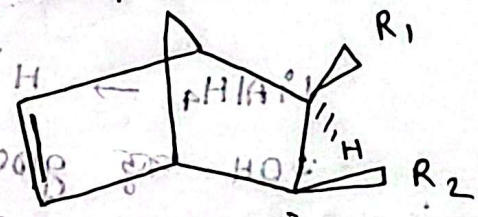
Diels Alder rxn approaches to the endo product following steneoselectivity :

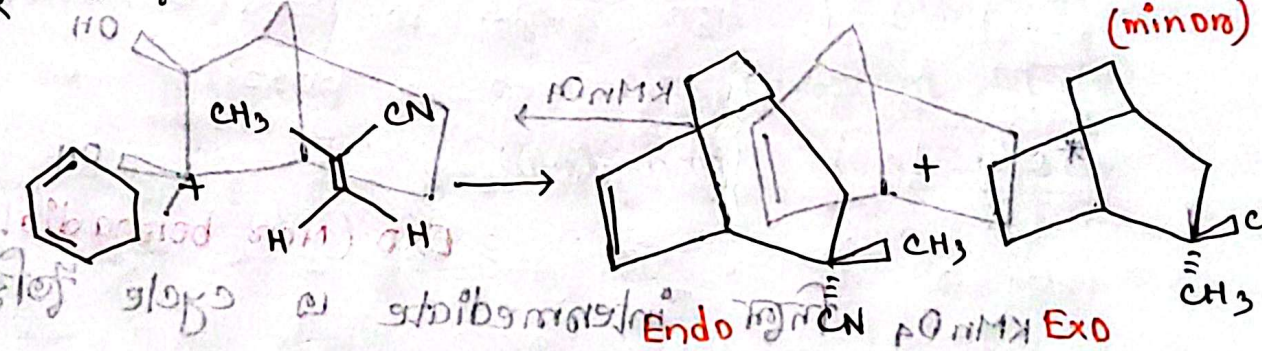
i) 

Product এর উপরে already $-CH_2-$ group থাকায় steric hindrance এর জন্য anhydride endo position-এ add হয় ।

ii) 

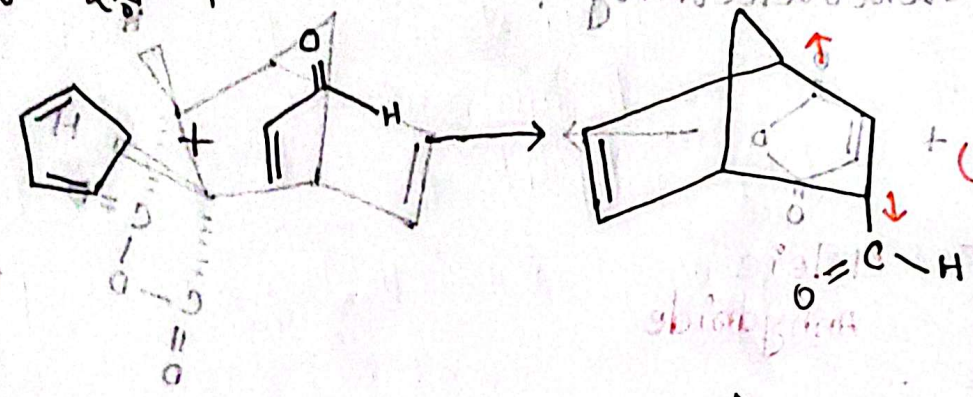
R_1, R_2 এর size বড় হলে endo-product হয় ।
কিন্তু ছোট হলে exo-endo দুটোই পাওয়া যায়, তখন endo major



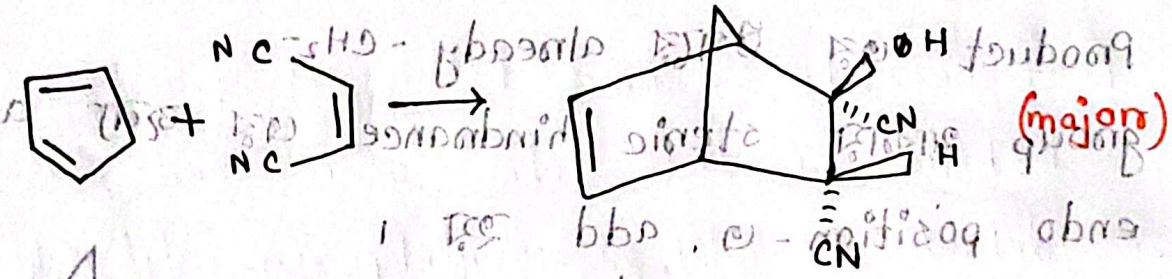
iii) 

e⁻-withdrawing group + বড় size হলে endo major হয়।

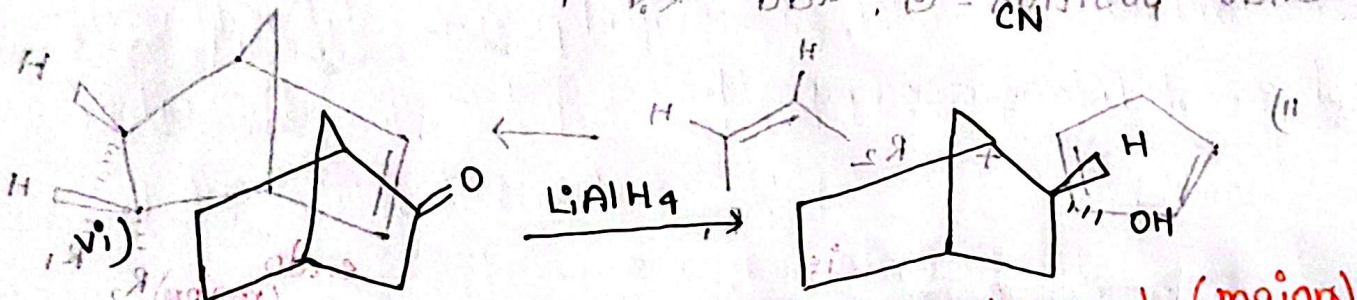
iv)



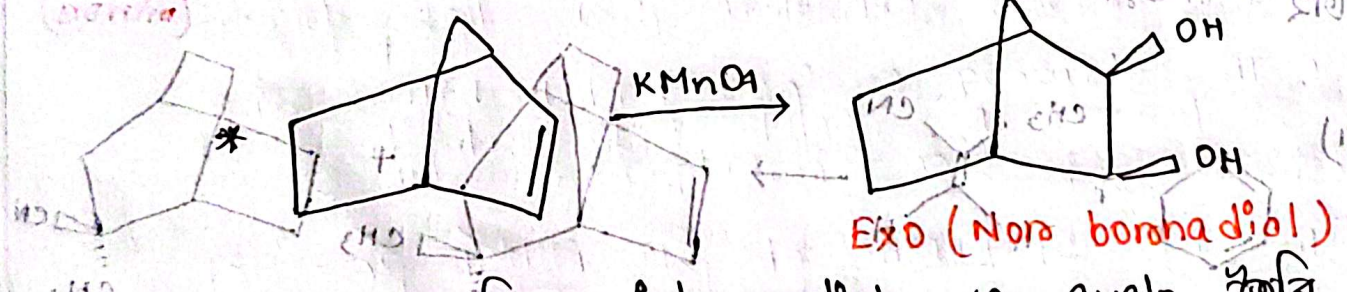
v)



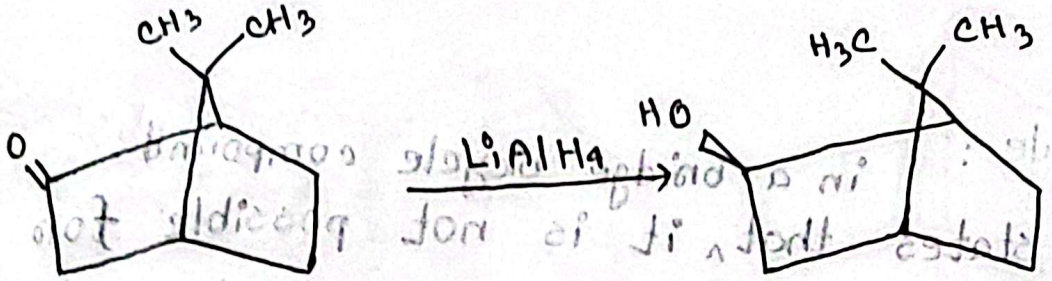
vi)



$LiAlH_4 \rightarrow H^-$ supply করবে Exo position দিয়ে। $-OH$ বড় group হওয়ায় endo-এ যাবে তাই endo product major



$KMnO_4$ দিলে intermediate এ cycle তৈরি হয় যা উপরে থাকলে steric hindrance হয়। তাই $KMnO_4$ endo approach করে কিন্তু product পাওয়া যাবে Exo

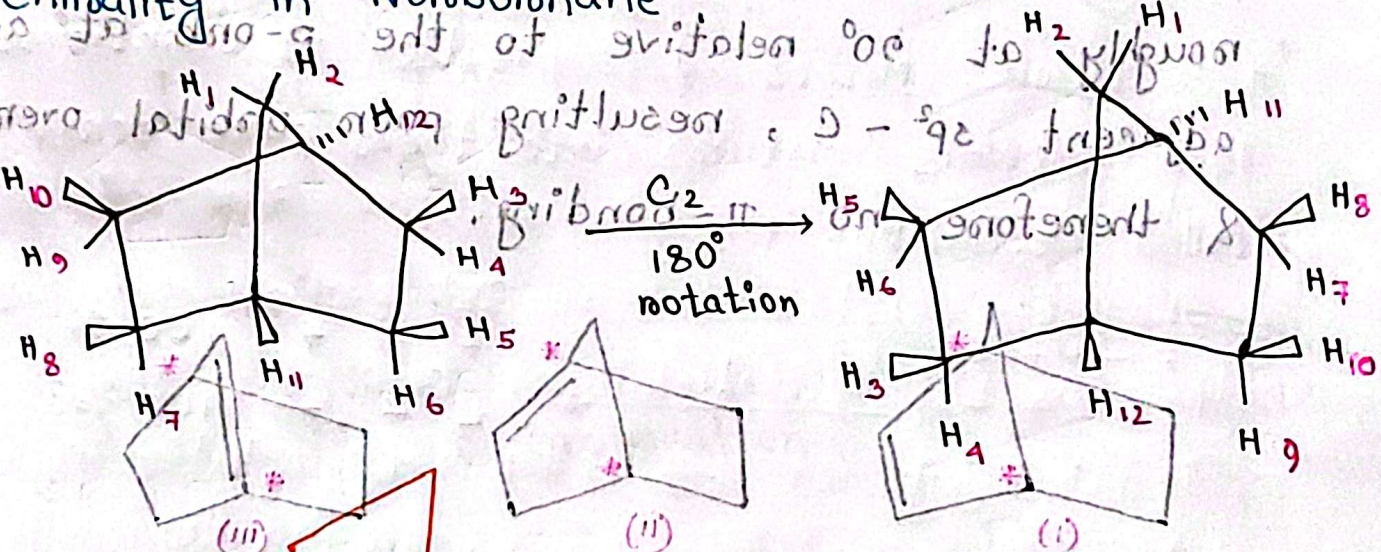


Exo - major

এখানে উপরে হটা methyl থাকায় H-addition exo দিয়ে হয় না। তাই H-addition → endo approach

কিন্তু Exo - product major.

* Chirality in Norbornane:



Norbornane এ axis of symmetry

এবং vertical plane of symmetry present থাকায়

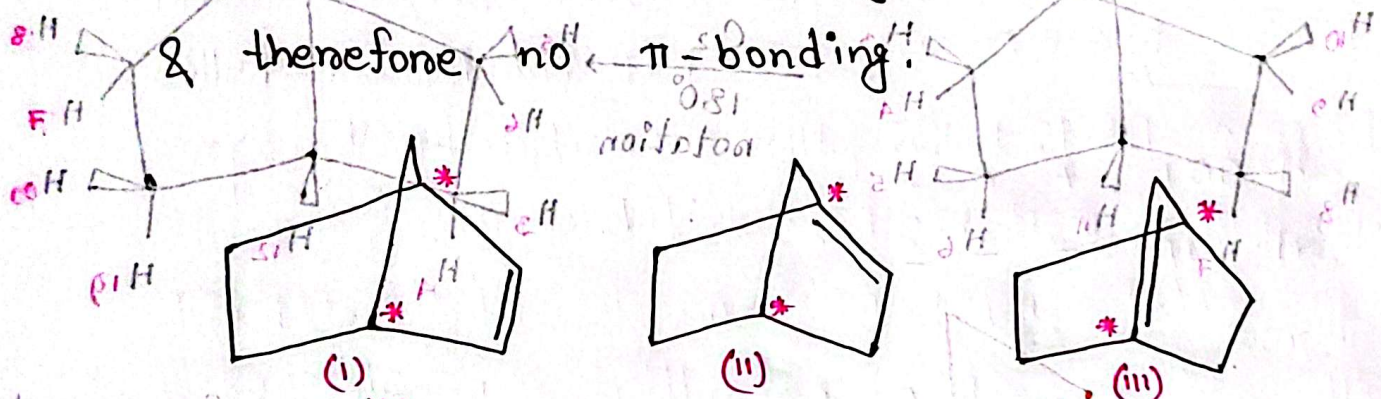
norbornane non-chiral.

C_{2v} Point Group

* Bredt's Rule : in a bridge bicycle compound

The rule states that, it is not possible for double bonds to be formed on the bridgehead C-atoms of a small bicyclic system if it is being incorporated in a ring compromised of fewer than 8 carbons.

The reason is that sp^2 carbons at bridgehead positions have a p-orbital that is oriented roughly at 90° relative to the p-orb at an adjacent sp^2 - C, resulting poor orbital overlap & therefore no π-bonding.

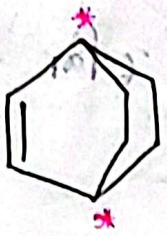


(I) (II) (III)

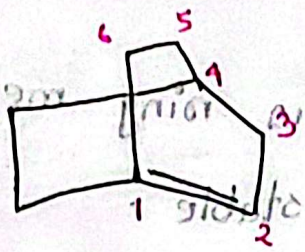
possible Not possible

p-orb গুলো 180° angle এ থাকলে π-bond form করতে পারে। কিন্তু 90° -এ possible না।

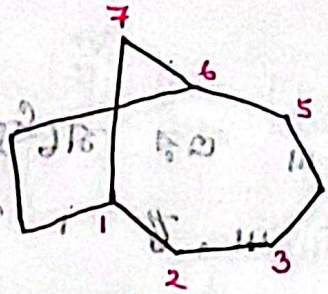
(II) ও (III) → non-planar, তাই p-orb গুলো 180° -এ নাই। তাই ~~do~~ পাশের C এর সাথে double bond form করতে পারে না। যদি ring-এর এক side এ 8+ C থাকে, তখন planar করা possible & তখন π-bond form হতে পারে।



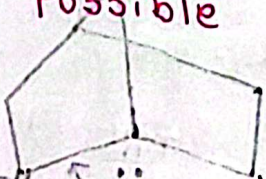
possible



Not-possible

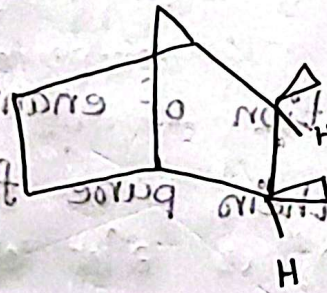


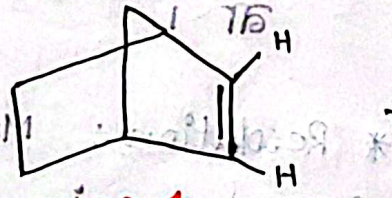


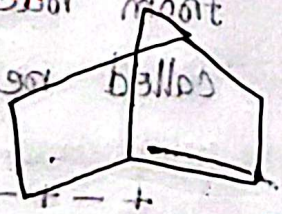


Possible

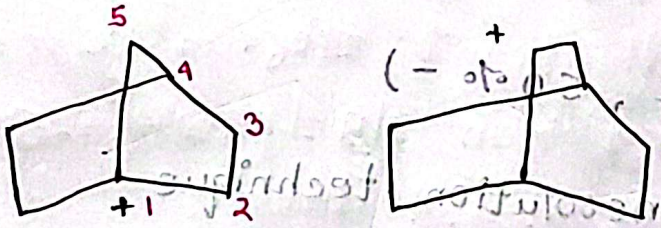
During β-elimination in bicyclic system, the bridgehead atoms are not involved.



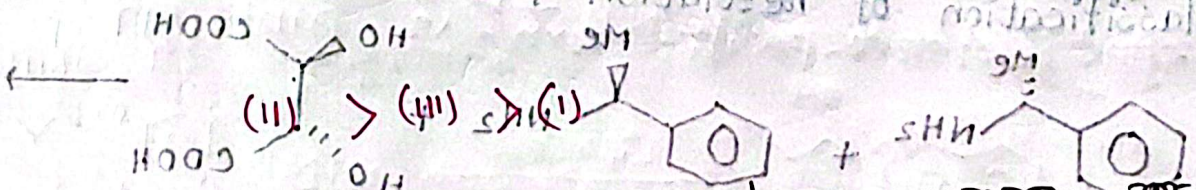




ii) Stability of carbocation:

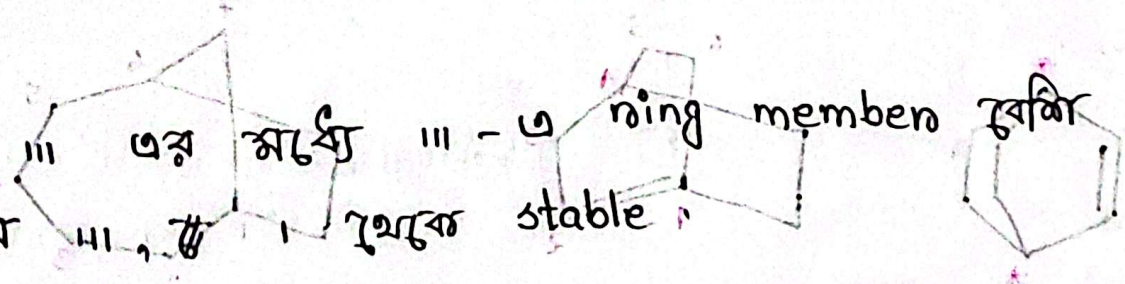


(i) (ii) (iii)

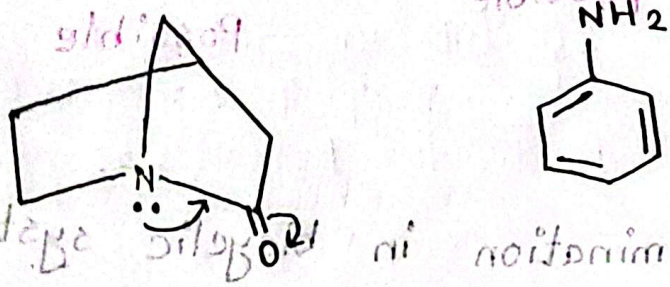


Bridge-carbon-এ +ve charge থাকলে তা হবে unstable. (ii)-এ charge bridge-এ এ নাই তাই এটা most stable.

I ও III এর মধ্যে III-এ ring member বেশি থাকায় III, II I থেকে stable

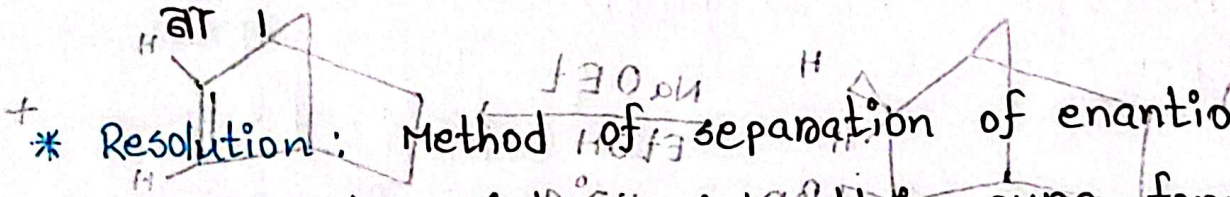


III)



এখানে bridgehead-এ double bond - possible না।

* Resolution: Method of separation of enantiomers from racemic mixture into their pure form is called resolution.



+ − + − + + + + 100%

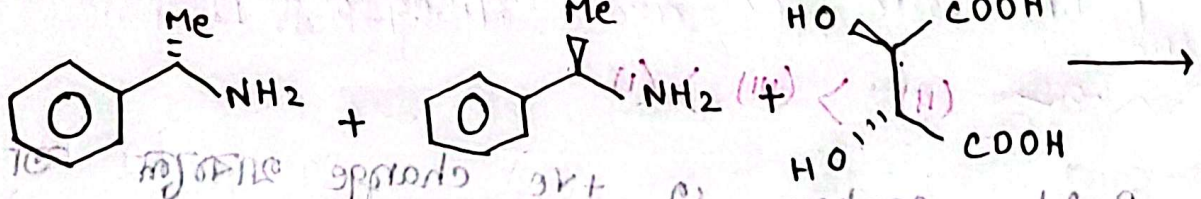
− + − + − resolution → − − 100%

+ − + − +

Racemic mixture (50% +, 50% −)

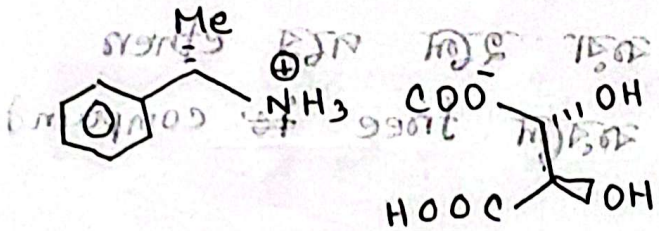
* Importance of resolution technique

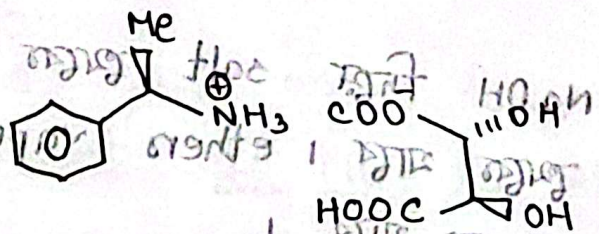
* Classification of resolution : 1. Chemical resolution



S(−) 1-phenylethyl amine

R,R (+) tartaric acid



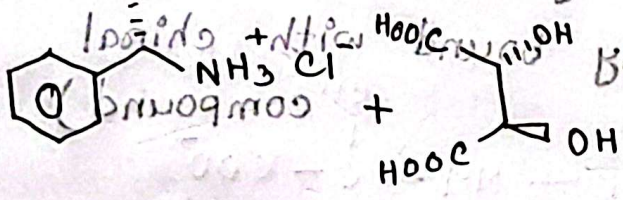


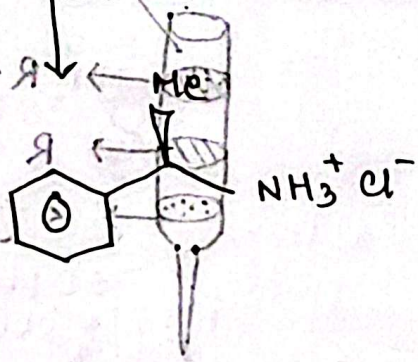
S(-) amine (+) T

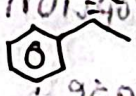
R (+) A (+) T

aq) HCl
extract^n with ether

aq HCl
extract^n with ether

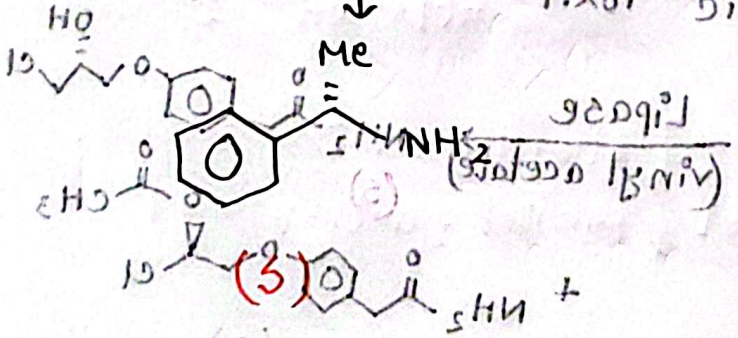


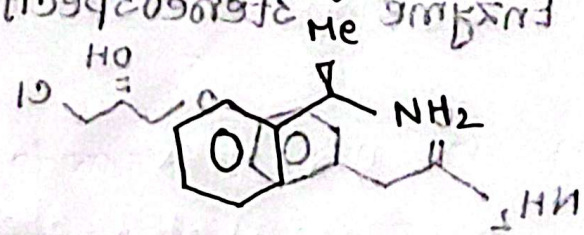


S(-) A(+) T এর salt form-এ আছে। তাকে soln-এ আনার জন্য aq HCl use করা হয়। HCl strong acid হওয়ায় $HCOO^-$ কে replace করে। প্রথমে  তৈরি হবে। extra HCl থাকলে product হবে পরে। carboxylic acid part (tantaric acid) ether এ soluble হবে। এর পর aq NaOH দিলে salt form থেকে আলাদা করা যায়।

aq NaOH
extract in ether

aq NaOH
extract in ether



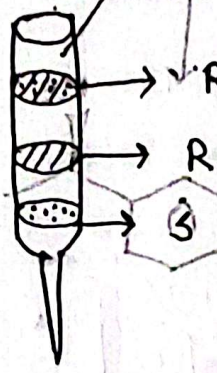


(S)

(R)

NaOH দিয়ে salt থেকে free করা হলে পরে ether থেকে যাবে। ether vaporize করলে free compound পাওয়া যাবে।

2. Chromatographic resolution :



→ chiral column packing (in stationary phase)

→ R + S (racemic mixture)

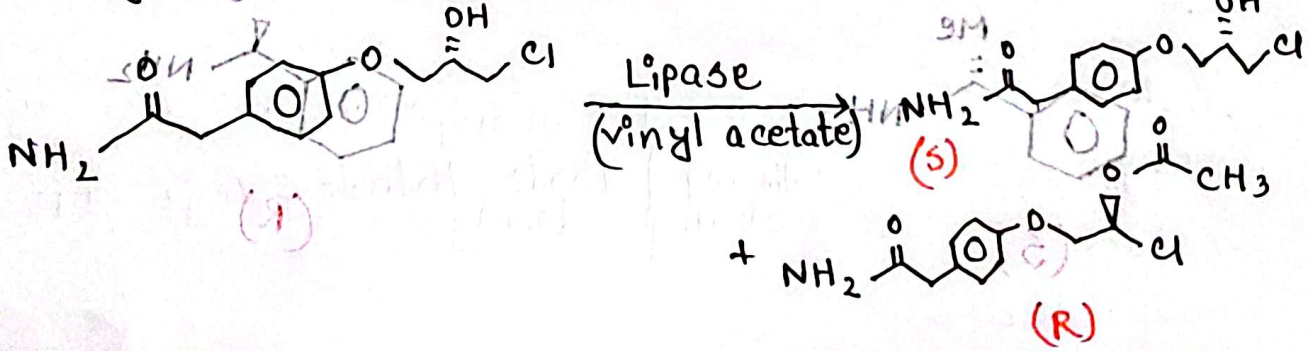
→ R (more tightly bound with chiral st. phase)

→ S (less tightly bound with chiral + compound)

Column Chromatography - এ SiO_2 এ chiral compound adsorption করানো হবে। chiral reagent stationary phase. R/S এর যেকোনো একটার affinity chiral reagent এর জন্য বেশি। সে stationary phase এর সাথে tightly bound থাকবে আর বাকিটা mobile phase এর সাথে run হয়ে আলাদা হয়ে যাবে।

3. Enzymatic resolution :

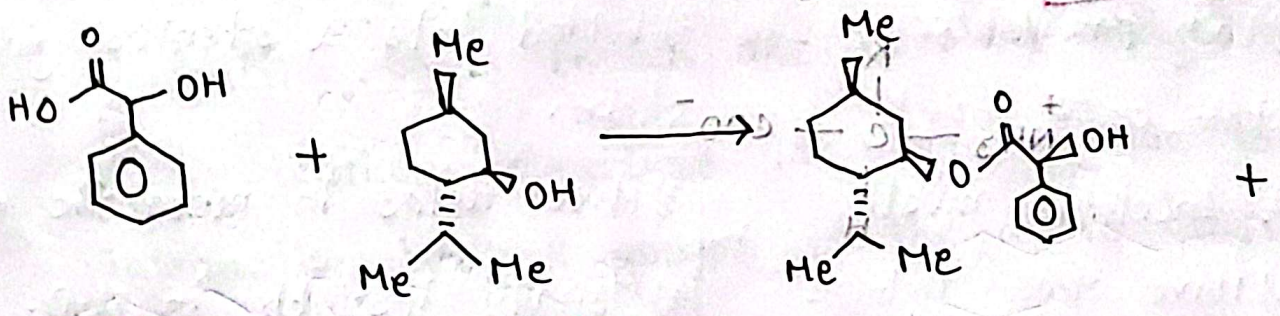
Enzyme stereospecific rxn - এ part নেয়।



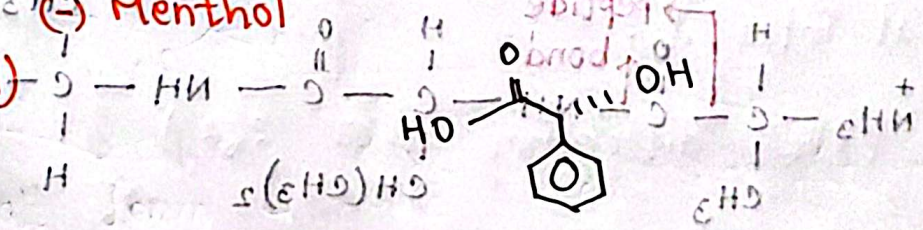
1° / 2° -COOH

Lipase enzyme 1° alcohol, এর সাথে esten form করে, কিন্তু steneospecific nxn হওয়ায় শুধু R - এর সাথে esten করবে। এভাবে আলাদা করা যায়।

1. Kinetic resolution :



R/S (Mandelic acid) (-) Menthol



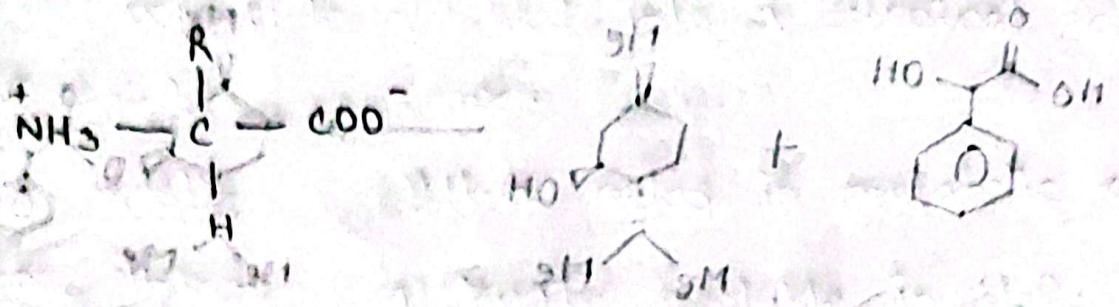
(-) Mandelic Acid

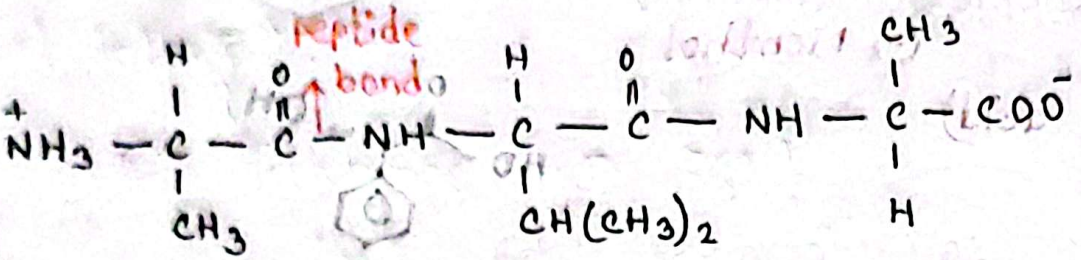
R/S এর যেকোনো ১টা fast & আরেকটা slow nxn করবে এভাবে আলাদা করা হয়।

Conformational Analysis of Biopolymers

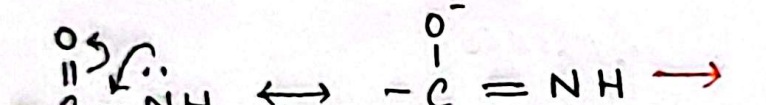
Biological polymer → carbohydrate, protein, fat molecule

* **Proteins** are polymers of: α-amino acids





The conformational flexibility of peptide chains is limited chiefly to rotation about the bonds leading to the α-C atoms. The C–N bond cannot rotate because of the rigidity of the amide bond. That's why peptide links are relatively planar & resistant to conformational analysis.



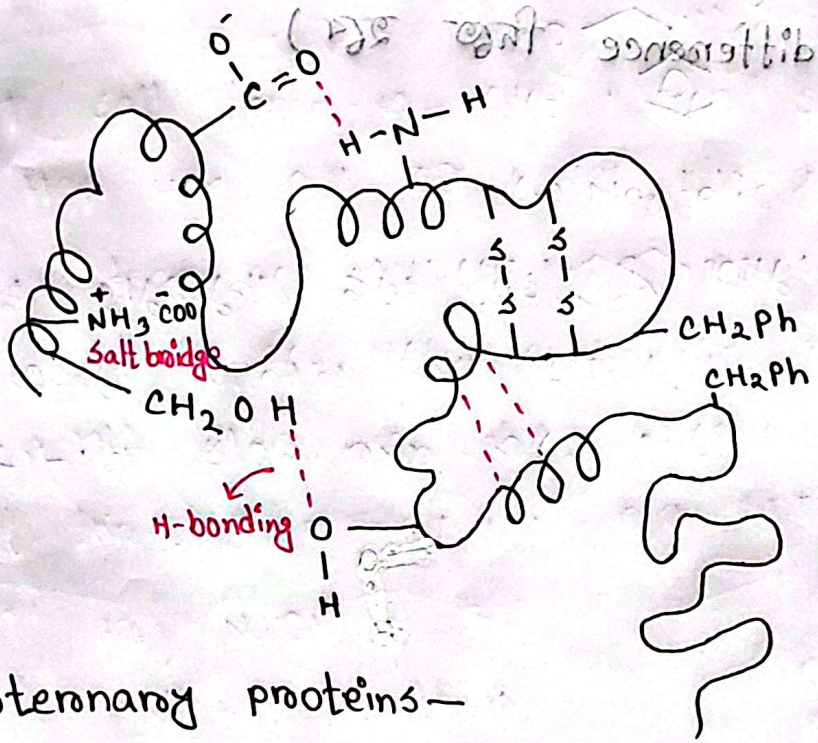
C = N bond এর জন্য rotation হয় না।

1) α - Helix
2) β-pleated sheet

} structure, Difference, parallel, anti-parallel

Tertiary structure of protein :

1) Non-polar hydrophobic interaction
2) Polar or hydrophilic
3) Salt bridge
4) Disulfide bonds (which are covalent bonds) from bet^n 2 SH of cystine molecule
5) The tertiary structure is primarily due to the interaction bet^n R groups of the amino acids.
6) H-bonding



Quarternary proteins—

1) Fibrous → α - keratine
2) Globular

Fibrous	Globular
1) Little on no tertiary structure.	1) A comple tertiary structure.
2) Long parallel polypeptide (chain on 2° chain)	2) Folded into a spherical globular shape.
3) Insoluble mostly	3) Have roles in metabolic rxn
4) Have cross linkage at intervals.	4) Usually insoluble in H_2O.

(আরো difference দিতে হবে)

